

Problem 4.11

The total amount of a loan to which interest has been added is \$20,000. The term of the loan was four and one-half years. If money accumulated at simple interest at a rate of 6%, what was the amount of the loan?

Solution

For simple interest,

$$A = P(1 + it).$$

Here,

$$A = 20000, \quad i = 6\% = 0.06, \quad t = 4.5.$$

So,

$$20000 = P(1 + 0.06 \cdot 4.5).$$

Thus,

$$20000 = P(1 + 0.27) = 1.27P.$$

Therefore,

$$P = \frac{20000}{1.27}.$$

Hence,

$$P \approx 15748.03.$$